Calcium absorption in Australian osteopenic post-menopausal women: an acute comparative study of fortified soymilk to cows' milk.

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Calcium loss after menopause increases the risk of osteoporosis in aging women. Soymilk is often consumed to reduce menopausal symptoms, although in its native form, it contains significantly less calcium than cow's milk. Moreover, when calcium is added as a fortificant, it may not be absorbed efficiently. This study compares calcium absorption from soymilk fortified with a proprietary phosphate of calcium versus absorption from cow's milk. Preliminary studies compared methods for labelling the calcium fortificant either before or after its addition to soymilk. It was established that fortificant labelled after it was added to soymilk had a tracer distribution pattern very similar to that shown by fortificant labelled before adding to soymilk, provided a heat treatment (90?C for 30 min) was applied. This method was therefore used for further bioavailability studies. Calcium absorption from fortified soy milk compared to cow's milk was examined using a randomised single-blind acute cross-over design study in 12 osteopenic post-menopausal women aged (mean +/- SD) 56.7+/-5.3 years, with a body mass index of 26.5+/-5.6 kg/m2. Participants consumed 20 mL of test milk labelled after addition of fortificant with 185 kBq of 45Ca in 44 mg of calcium carrier, allowing the determination of the hourly fractional calcium absorption rate (alpha) using a single isotope radiocalcium test. The mean hourly fractional calcium absorption from fortified soymilk was found to be comparable to that of cows' milk: alpha = 0.65+/-0.19 and alpha =0.66+/-0.22, p>0.05, respectively.